

Project Title:

Smart Building Security, Access Control and IoT Monitoring System

Overview of The Project:

- Imagine a modern software company where only authorized employees are allowed to enter the building. To ensure the safety of employees, valuable equipment, and confidential information, the company installs the **Smart Building Security, Access Control and IoT Monitoring System** at the main entrance. The entire system is controlled by the **LPC2129 ARM7 microcontroller**, which acts as the brain of the project. Every sensor, communication module, storage device, and output device is connected to the LPC2129, allowing it to monitor and control the complete security system.
- As soon as a person approaches the entrance, the **PIR Motion Sensor** detects human movement near the door. This indicates that someone is waiting to enter the building. The LPC2129 immediately becomes active and prepares the authentication system. At the

same time, the LCD welcomes the visitor and asks them to begin the authentication process.

- The first level of authentication is performed using the **RC522 RFID Reader**. The employee places an RFID card near the reader. The RC522 reads the unique identification number stored inside the card and sends it to the LPC2129 through the SPI interface. The LPC2129 compares the received card number with the list of authorized users stored in the system. If the RFID card is valid, the authentication process is completed immediately and the employee is allowed to enter the building.
- If the RFID card is not recognized or the employee does not have the card, the system automatically switches to the second level of authentication, which is the **R307S Fingerprint Sensor**. Since fingerprint processing requires several commands and template matching operations, the fingerprint sensor is connected to an **Arduino UNO**. The Arduino captures the fingerprint, compares it with the stored fingerprint database, and sends only the final result such as **Fingerprint Matched** or **Fingerprint Not Matched** to the LPC2129 through UART communication. This keeps the LPC2129 focused on overall system control while the Arduino performs the fingerprint processing.

- If both the RFID card and fingerprint authentication fail, the system provides a third level of authentication using the **4×4 Matrix Keypad**. The user enters the password through the keypad. The LPC2129 compares the entered password with the stored password. If the password is correct, the user is authenticated successfully. If the password is incorrect, the system counts the failed attempts. After multiple incorrect attempts, the LPC2129 considers it an unauthorized access attempt and activates the security alarm.
- Once the user is successfully authenticated, the LPC2129 begins the door access sequence. It energizes the **Relay Module**, which supplies power to the **Electromagnetic Door Lock**. The lock is released, allowing the door to be opened. Simultaneously, the LPC2129 controls the **Servo Motor**, which automatically opens the door for a few seconds and then closes it again after the user has entered the building. This provides a completely automatic access control mechanism without requiring manual door operation.
- Throughout the authentication process, the **16×2 LCD Display** continuously provides useful information to the user. Messages such as "**Scan RFID Card**," "**Fingerprint Verification**," "**Enter Password**," "**Access Granted**," "**Access Denied**," and "**Door**

Open" are displayed so that the user clearly understands the current status of the system. The LCD acts as the primary user interface for the entire project.

- The project also provides visual and audible indications using **LEDs** and a **Buzzer**. During successful authentication, the Green LED is turned ON to indicate that access has been granted. During authentication, the Yellow LED may indicate that verification is in progress. Whenever unauthorized access is detected, the Red LED turns ON and the Buzzer produces an alarm sound to alert nearby security personnel.
- Every security event is recorded with the help of the **DS1307 Real Time Clock (RTC)**. The RTC continuously provides the current date and time to the LPC2129. Whenever a user enters the building, fails authentication, or triggers an alarm, the exact time and date are attached to the event. This creates an accurate record of every activity occurring within the building.
- The project stores important information using two different storage devices. The **AT24C256 EEPROM** stores permanent system information such as configuration parameters, passwords, and authorized user data that must be retained even after power failure. The **SD Card Module** stores detailed event

logs, including user ID, authentication method, entry time, exit time, and alarm history. These logs can later be reviewed by the administrator for security analysis and auditing.

- For remote monitoring, the **ESP-01 Wi-Fi Module** connects the LPC2129 to the Internet. Important events such as successful entry, failed authentication, alarm activation, and door status are uploaded to the cloud. This enables the building administrator to monitor the complete security system remotely from anywhere using an Internet connection.
- The project also supports communication with multiple security controllers using the **MCP2551 CAN Transceiver**. If the building contains multiple entrances or different security zones, each entrance can have its own LPC2129 controller. Whenever an important security event occurs at one entrance, the CAN network immediately shares this information with all other LPC2129 controllers, allowing the entire building security system to operate as one integrated network.
- Finally, the **Magnetic Door Sensor** continuously monitors whether the door is properly closed after every operation. If the door is forced open without successful authentication or remains open for an unusually long time, the LPC2129 immediately detects

the abnormal condition. It activates the buzzer, turns ON the Red LED, records the event in the SD card, stores the time using the RTC, uploads the alert to the cloud through the ESP-01, and broadcasts the warning to other controllers through the CAN network.

- In this way, every module in the project works together under the control of the LPC2129 ARM7 microcontroller to provide a **secure, intelligent, automatic, and IoT-enabled building access control system**. The project not only authenticates users but also controls the door, records every activity, monitors security conditions, communicates with other controllers, and provides remote supervision, making it a complete real-world embedded security solution suitable for offices, industries, laboratories, apartments, educational institutions, and commercial buildings.

Technology Used:

- LPC2129 ARM7 Microcontroller
- Embedded C
- RC522 RFID Reader
- R307S Fingerprint Sensor
- Arduino UNO
- 4×4 Matrix Keypad
- PIR Motion Sensor

- Magnetic Door Sensor
- DS1307 RTC
- ESP-01 Wi-Fi Module
- MCP2551 CAN Transceiver
- AT24C256 EEPROM
- SD Card Module
- Relay Module
- Electromagnetic Door Lock
- Servo Motor
- 16×2 LCD
- LEDs
- Buzzer
- ThingSpeak IoT Cloud

Project Explanation

The Smart Building Security, Access Control and IoT Monitoring System is an intelligent embedded security solution developed using the LPC2129 ARM7 Microcontroller. The system authenticates users using RFID cards, fingerprint verification through an Arduino-controlled R307S sensor, and password authentication through a keypad. After successful verification, the controller unlocks the electromagnetic door lock using a relay and servo motor. PIR and magnetic door sensors continuously monitor building security, while the RTC records every event with date and time. EEPROM and SD card store security logs, the ESP-01 uploads data to the ThingSpeak cloud, and the MCP2551 CAN transceiver exchanges information with other security nodes. The system provides

secure, reliable, and scalable access control for modern smart buildings.

Introduction

Modern buildings require intelligent security systems that can authenticate users, monitor activity, record events, and support remote monitoring. Conventional lock-and-key systems cannot maintain entry logs or provide cloud connectivity. This project integrates embedded systems, IoT, industrial communication, and multiple authentication methods into one secure platform using the LPC2129 ARM7 microcontroller.

Objective of the Project

- Provide secure building access using multiple authentication methods.
- Prevent unauthorized entry.
- Automatically control the electromagnetic door lock.
- Monitor human movement and door status.
- Record events with date and time.
- Store security records in EEPROM and SD card.
- Support cloud monitoring using ESP-01.
- Enable CAN communication between multiple nodes.

Need of the Project

Traditional security systems provide limited protection and require manual operation. They cannot verify user identity, maintain complete event logs, or provide remote monitoring. This project overcomes these limitations by integrating RFID, fingerprint, password authentication, event logging, cloud monitoring, CAN networking, and automatic access control into one embedded security system.

Proposed System

The proposed system uses the LPC2129 ARM7 Microcontroller as the central controller. The RC522 RFID Reader performs initial authentication, followed by fingerprint verification using the Arduino-controlled R307S sensor. If required, password authentication is performed through the keypad. After successful authentication, the relay energizes the electromagnetic door lock and the servo motor opens the door. PIR and magnetic door sensors continuously monitor security conditions. The DS1307 RTC records the event time, the AT24C256 EEPROM and SD card store event logs, the ESP-01 uploads information to the ThingSpeak cloud, and the MCP2551 CAN transceiver exchanges data with other LPC2129 security nodes.

Features of the Proposed System

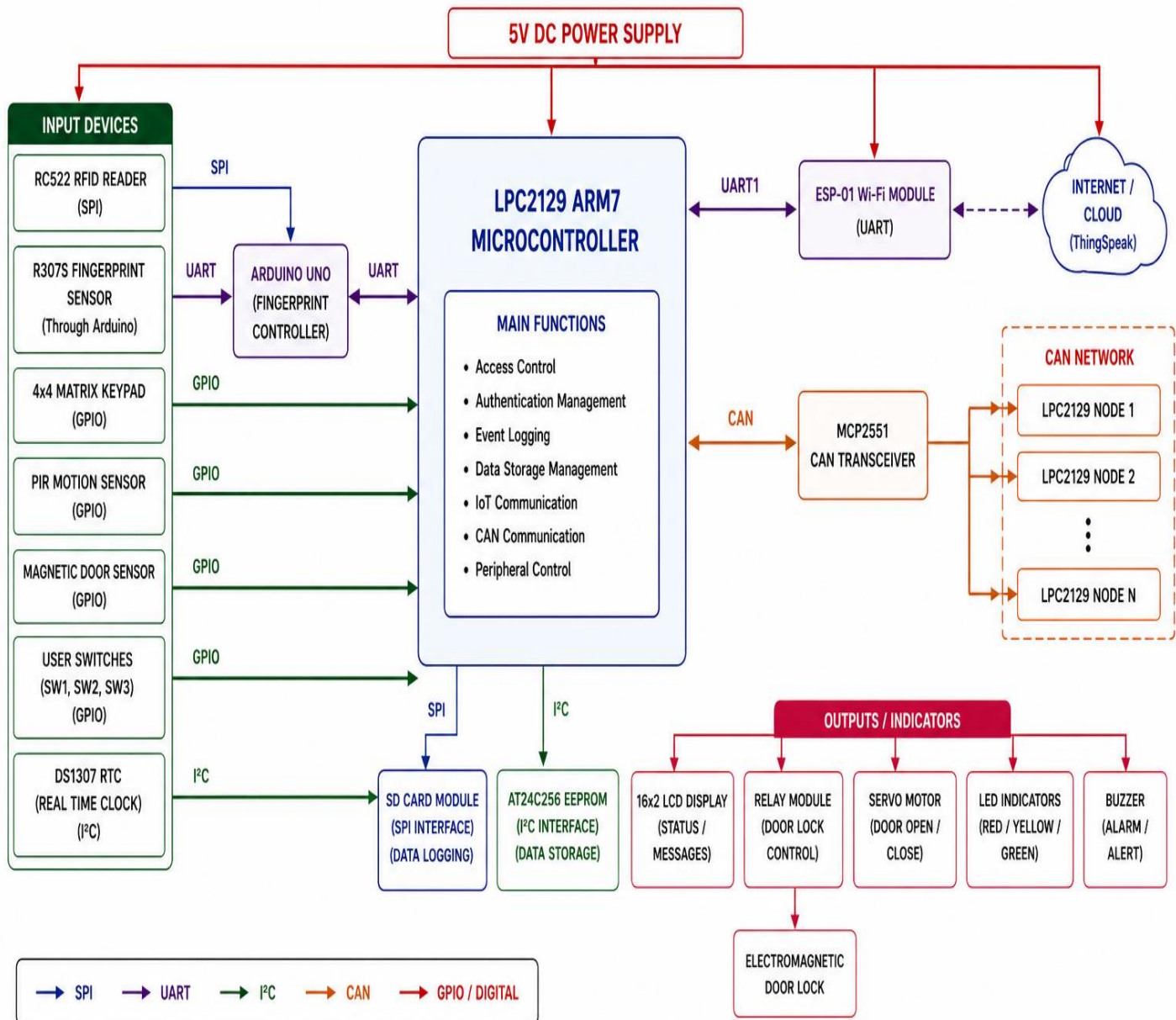
- Multi-level authentication using RFID, Fingerprint and Password.
- Automatic door locking and unlocking.
- PIR-based motion detection.
- Magnetic door status monitoring.
- Cloud monitoring through ESP-01 and ThingSpeak.
- CAN communication between multiple security nodes.
- Event logging using RTC, EEPROM and SD Card.
- LCD, LEDs and buzzer for user indication.
- Scalable architecture for offices, industries and smart buildings.

Main Concept of the Project

The main concept of this project is to provide secure building access by combining multiple authentication methods with continuous security monitoring and IoT connectivity. The LPC2129 ARM7 Microcontroller coordinates authentication, door control, sensing, data logging, cloud communication and CAN networking. Only authorized users are allowed to access the building, while every event is recorded for future analysis.

Block Diagram:

SMART BUILDING SECURITY, ACCESS CONTROL AND IoT MONITORING SYSTEM



Working Principle

When a user approaches the entrance, the RC522 RFID Reader first verifies the RFID card. If additional verification is required, the Arduino-controlled R307S Fingerprint Sensor authenticates the user. Password authentication through the 4×4 keypad can also be performed whenever necessary. After successful verification, the LPC2129 energizes the relay to release the electromagnetic door lock and drives the servo motor to open the door. PIR and magnetic door sensors continuously monitor security conditions. Every event is recorded using the DS1307 RTC, stored in the EEPROM and SD Card, uploaded to the ThingSpeak cloud through the ESP-01 Wi-Fi module and shared with other LPC2129 nodes through the MCP2551 CAN transceiver.

System Architecture

The LPC2129 ARM7 Microcontroller acts as the central controller of the complete security system. Authentication devices provide user credentials, sensors monitor building security, storage devices maintain event logs, communication modules exchange information with the cloud and CAN network, and output devices provide user notification and access control. This modular architecture improves reliability, scalability and ease of maintenance.

Hardware Description

The detailed Hardware Description section will be prepared in the same style as Project-1. Each hardware module from the approved Project-2 block diagram will include:

- Explanation
- UART Output
- LCD Output
- LED Indication

Modules covered:

1. LPC2129 ARM7
2. RC522 RFID Reader

RC522 RFID Reader

The **RC522 RFID Reader** is the primary authentication device used in the **Smart Building Security, Access Control and IoT Monitoring System**. It communicates with the **LPC2129 ARM7 Microcontroller** through the **SPI (Serial Peripheral Interface)** protocol and is responsible for reading the unique identification (UID) stored inside an RFID card or tag. When a user brings the RFID card near the reader, the RC522 captures the card information and sends it to the LPC2129 for authentication.

The LPC2129 compares the received RFID UID with the authorized IDs stored in the system. If the RFID card is valid,

the authentication process is successful, and the controller proceeds with the next level of verification (Fingerprint Authentication) or directly grants access based on the configured security level. If the RFID card is not authorized, access is denied, an alert message is displayed, and the failed authentication attempt is recorded in the EEPROM and SD Card with the current date and time obtained from the RTC.

The RC522 RFID Reader provides fast, contactless, and reliable authentication, reducing manual effort while improving building security. It is widely used in offices, industries, apartments, laboratories, educational institutions, and other secure access control systems.

UART Output

Case-1: RFID Reader Initialization

RC522 RFID Reader Initialized

SPI Communication Successful

Waiting for RFID Card...

Case-2: Authorized RFID Card

RFID Card Detected

Card ID : A1 B2 C3 D4

RFID Authentication Successful

Proceeding to Fingerprint Verification...

Case-3: Unauthorized RFID Card

RFID Card Detected

Card ID : 8F 2A 91 C5

RFID Authentication Failed

Access Denied

Event Logged Successfully

LCD Output

During Initialization

RFID Ready

Scan Card

Authorized Card

RFID Success

Verify Finger

Unauthorized Card

Invalid Card

Access Denied

LED Indication

RFID Reader Ready

 Green LED : ON

 Yellow LED : OFF

 Red LED : OFF

Authorized RFID Card

 Green LED : ON

 Yellow LED : OFF

 Red LED : OFF

Unauthorized RFID Card

 Green LED : OFF

 Yellow LED : OFF

 Red LED : ON

R307S Fingerprint Sensor

Explanation

The **R307S Fingerprint Sensor** is used to provide **biometric authentication** in the Smart Building Security, Access Control and IoT Monitoring System. After the **RC522 RFID Reader** successfully verifies the RFID card, the **Arduino UNO** activates the R307S Fingerprint Sensor. The sensor verifies the user's fingerprint and sends the authentication result to the **LPC2129 ARM7 Microcontroller** through the Arduino UNO. If the

fingerprint is valid, the LPC2129 unlocks the door using the **Relay Module** and **Electromagnetic Door Lock**, operates the **Servo Motor**, records the event in the **EEPROM** and **SD Card**, and uploads the information to the cloud using the **ESP-01 Wi-Fi Module**. If the fingerprint is invalid, access is denied and the buzzer is activated.

Expected UART Output

Initialization

Fingerprint Sensor Initialized

Waiting for Finger...

Valid Fingerprint

Fingerprint Detected

Fingerprint Verified

Access Granted

Door Opening...

Invalid Fingerprint

Fingerprint Detected

Fingerprint Not Matched

Access Denied

Alarm Activated

Expected LCD Output

Initialization

Fingerprint Ready

Place Finger

Valid Fingerprint

Finger Verified

Door Opening

Invalid Fingerprint

Invalid Finger

Access Denied

LED Indication

Fingerprint Verified

 Green LED : ON

 Red LED : OFF

Fingerprint Failed

 Green LED : OFF

 Red LED : ON

4×4 Matrix Keypad

Explanation

The **4×4 Matrix Keypad** is used for **password-based authentication** in the Smart Building Security, Access Control and IoT Monitoring System. It is connected to the **LPC2129 ARM7 Microcontroller** through GPIO pins. After successful RFID authentication, the system can ask the user to enter a password using the keypad. The LPC2129 compares the entered password with the stored password. If the password is correct, the authentication process continues to the next step. If the password is incorrect, access is denied and the security event is recorded.

Expected UART Output

Initialization

4x4 Keypad Initialized

Waiting for Password...

Correct Password

Password Entered : ****

Password Verified

Proceeding to Fingerprint Verification...

Wrong Password

Password Entered : ****

Invalid Password

Access Denied

Alarm Activated

Expected LCD Output

Initialization

Enter Password

Correct Password

Password OK

Verify Finger

Wrong Password

Wrong Password

Access Denied

LED Indication

Correct Password

 Green LED : ON

 Red LED : OFF

Wrong Password

 Green LED : OFF

 Red LED : ON

PIR Motion Sensor

Explanation

The **PIR (Passive Infrared) Motion Sensor** is used to detect **human movement** near the building entrance. It is connected to the **LPC2129 ARM7 Microcontroller** through a GPIO pin. When motion is detected, the LPC2129 records the event with the current **date and time** using the **DS1307 RTC**, stores it in the **EEPROM** and **SD Card**, and uploads the information to the **ThingSpeak Cloud** through the **ESP-01 Wi-Fi Module**. If motion is detected without successful user authentication, the system activates the security alarm.

Expected UART Output

Initialization

PIR Motion Sensor Initialized

Monitoring for Motion...

Motion Detected

Motion Detected

Checking User Authentication...

Event Logged Successfully

Unauthorized Motion

Unauthorized Motion Detected

Alarm Activated

Security Alert Sent

Expected LCD Output

Initialization

PIR Ready

Monitoring...

Motion Detected

Motion Detected

Checking User

Unauthorized Motion

Security Alert

Access Denied

LED Indication

No Motion

 Green LED : ON

 Red LED : OFF

Motion Detected

 Green LED : Blinking

 Red LED : OFF

Unauthorized Motion

 Green LED : OFF

 Red LED : ON

 Buzzer : ON

Magnetic Door Sensor

Explanation

The **Magnetic Door Sensor** is used to monitor the **door status** in the Smart Building Security, Access Control and IoT Monitoring System. It is connected to the **LPC2129 ARM7 Microcontroller** through a GPIO pin. The sensor detects whether the door is **Open** or **Closed**. Whenever the door status changes, the LPC2129 records the event using the **DS1307 RTC**, stores it in the **EEPROM** and **SD Card**, and uploads the information to the **ThingSpeak Cloud** through the **ESP-01 Wi-Fi Module**. If the door is opened without successful authentication, the system activates the security alarm.

Expected UART Output

Initialization

Magnetic Door Sensor Initialized

Monitoring Door Status...

Door Opened (Authorized)

Door Opened

Access Authorized

Event Logged Successfully

Door Closed

Door Closed

Door Locked

System Secured

Door Opened (Unauthorized)

Unauthorized Door Opening

Alarm Activated

Security Alert Sent

Expected LCD Output

Initialization

Door Sensor Ready

Monitoring...

Door Opened

Door Opened

Welcome

Door Closed

Door Closed

Locked

Unauthorized Door Opening

Security Alert

Door Breach

LED Indication

Door Closed (Normal Condition)

 Green LED : ON

 Red LED : OFF

Door Opened (Authorized)

 Green LED : Blinking

 Red LED : OFF

Unauthorized Door Opening

 Green LED : OFF

 Red LED : ON

 **Buzzer : ON**

User Switches (SW1, SW2, SW3)

Explanation

The **User Switches (SW1, SW2, and SW3)** are used for **manual operation and system testing** in the Smart Building Security, Access Control and IoT Monitoring System. They are connected to the **LPC2129 ARM7 Microcontroller** through GPIO pins. Depending on the program, the switches can be used to **unlock the door manually, activate emergency mode, reset alarms, or test system functions**. Whenever a

switch is pressed, the LPC2129 performs the corresponding operation and records the event for monitoring.

Expected UART Output

Initialization

User Switches Initialized

Waiting for User Input...

SW1 Pressed (Manual Door Unlock)

SW1 Pressed

Manual Door Unlock

Door Opened

SW2 Pressed (Alarm Reset)

SW2 Pressed

Security Alarm Reset

System Ready

SW3 Pressed (Emergency Mode)

SW3 Pressed

Emergency Mode Activated

Door Unlocked

Expected LCD Output

Initialization

Switches Ready

Waiting...

SW1 Pressed

Manual Access

Door Opened

SW2 Pressed

Alarm Reset

System Ready

SW3 Pressed

Emergency Mode

Activated

LED Indication

Normal Condition

 Green LED : ON

 Red LED : OFF

SW1 Pressed

 Green LED : Blinking

 Red LED : OFF

SW2 Pressed

 Green LED : ON

 Red LED : OFF

SW3 Pressed

 Green LED : Blinking

 Red LED : ON

DS1307 RTC (Real Time Clock)

Explanation

The **DS1307 Real Time Clock (RTC)** is used to maintain the **current date and time** in the Smart Building Security, Access Control and IoT Monitoring System. It communicates with the **LPC2129 ARM7 Microcontroller** through the **I²C protocol**. Whenever an important event occurs, such as **RFID authentication, fingerprint verification, door opening, motion detection, or unauthorized access**, the RTC provides the current date and time. The LPC2129 stores this information in the **EEPROM, SD Card**, and uploads it to the **ThingSpeak Cloud** for accurate event tracking.

Expected UART Output

Initialization

DS1307 RTC Initialized

Current Time Loaded

RTC Running...

Event Logged

Date : 30/07/2026

Time : 10:45:18

Security Event Logged

RTC Communication Error

RTC Communication Failed

Check I2C Connection

Expected LCD Output

Initialization

RTC Ready

Time Loaded

Normal Operation

30/07/2026

10:45:18

RTC Error

RTC Error

Check RTC

LED Indication

RTC Working Normally

 Green LED : ON

 Red LED : OFF

RTC Communication Error

 Green LED : OFF

 Red LED : ON

ESP-01 Wi-Fi Module

Explanation

The **ESP-01 Wi-Fi Module** is used to provide **IoT connectivity** in the Smart Building Security, Access Control and IoT Monitoring System. It communicates with the **LPC2129 ARM7 Microcontroller** through **UART communication**. The ESP-01 uploads important security events such as **RFID authentication, fingerprint verification, door status, motion detection, and security alerts** to the **ThingSpeak Cloud**. This allows the administrator to monitor the building remotely through the Internet.

Expected UART Output

Initialization

ESP-01 Initialized

Connecting to Wi-Fi...

Wi-Fi Connected

ThingSpeak Connected

Data Uploaded Successfully

Uploading Security Data...

Upload Successful

Cloud Updated

Wi-Fi Connection Failed

Connecting to Wi-Fi...

Connection Failed

Retrying...

Expected LCD Output

Initialization

Wi-Fi Ready

Connected

Data Upload

Uploading...

Success

Connection Failed

Wi-Fi Error

Retry...

LED Indication

Wi-Fi Connected

 Green LED : ON

 Red LED : OFF

Uploading Data

 Green LED : Blinking

 Red LED : OFF

Wi-Fi Connection Failed

 Green LED : OFF

 Red LED : ON

AT24C256 EEPROM

Explanation

The **AT24C256 EEPROM** is used to store **important security data permanently** in the Smart Building Security, Access Control and IoT Monitoring System. It communicates with the **LPC2129 ARM7 Microcontroller** through the **I²C protocol**. The EEPROM stores information such as **authorized user IDs, system settings, and important security events**. Since EEPROM is **non-volatile memory**, the stored data is retained even after the power is turned OFF.

Expected UART Output

Initialization

AT24C256 EEPROM Initialized

I2C Communication Successful

EEPROM Ready

Data Stored Successfully

Saving Security Data...

Data Stored Successfully

EEPROM Write Complete

Data Read Successfully

Reading Security Data...

Data Read Successfully

EEPROM Error

EEPROM Communication Failed

Check I2C Connection

Expected LCD Output

Initialization

EEPROM Ready

Data Stored

Data Saved

Data Read

Data Loaded

EEPROM Error

EEPROM Error

LED Indication

EEPROM Working Normally

 Green LED : ON

 Red LED : OFF

Data Writing/Reading

 Green LED : Blinking

 Red LED : OFF

EEPROM Communication Error

 Green LED : OFF

 Red LED : ON

SD Card Module

Explanation

The **SD Card Module** is used to store **security event logs** in the Smart Building Security, Access Control and IoT Monitoring System. It communicates with the **LPC2129 ARM7 Microcontroller** through the **SPI protocol**. The SD Card stores

important information such as **RFID authentication, fingerprint verification, door status, motion detection, and security alerts** with the **date and time** received from the **DS1307 RTC**. This data can be accessed later for monitoring, reporting, and security analysis.

Expected UART Output

Initialization

SD Card Initialized

SPI Communication Successful

SD Card Ready

Data Stored Successfully

Saving Security Log...

Log Saved Successfully

SD Card Write Complete

Data Read Successfully

Reading Security Log...

Log Read Successfully

SD Card Error

SD Card Not Detected

Storage Failed

Check SD Card

Expected LCD Output

Initialization

SD Card Ready

Data Stored

Log Saved

Data Read

Log Loaded

SD Card Error

SD Card Error

LED Indication

SD Card Ready

 Green LED : ON

 Red LED : OFF

Saving/Reading Data

 Green LED : Blinking

 Red LED : OFF

SD Card Error

 Green LED : OFF

 Red LED : ON

MCP2551 CAN Transceiver

Explanation

The **MCP2551 CAN Transceiver** is used to provide **CAN (Controller Area Network) communication** between multiple **LPC2129 ARM7-based security systems**. It acts as an interface between the **LPC2129 CAN Controller** and the **CAN Bus**. In this project, the MCP2551 allows multiple building security nodes to exchange information such as **RFID authentication status, fingerprint verification, door status, motion detection, and security alerts**. This enables centralized monitoring and improves the security of large buildings with multiple entrances.

Expected UART Output

Initialization

MCP2551 Initialized

CAN Communication Ready

Waiting for CAN Message...

CAN Message Transmitted

Sending Security Status...

CAN Message Sent

Transmission Successful

CAN Message Received

CAN Message Received

Node ID : 02

Door Status : Open

CAN Communication Error

CAN Communication Failed

Retrying...

Expected LCD Output

Initialization

CAN Ready

Message Transmitted

CAN Tx OK

Message Received

CAN Rx OK

CAN Error

CAN Error

LED Indication

CAN Network Ready

 Green LED : ON

 Red LED : OFF

CAN Data Transmission/Reception

 Green LED : Blinking

 Red LED : OFF

CAN Communication Error

 Green LED : OFF

 Red LED : ON

16×2 LCD Display

Explanation

The **16×2 LCD Display** is used to display the **system status and user messages** in the Smart Building Security, Access Control and IoT Monitoring System. It is connected to the

LPC2129 ARM7 Microcontroller. The LCD provides real-time information such as **RFID authentication, fingerprint verification, password entry, door status, motion detection, Wi-Fi status, and security alerts**, making the system easy to operate and monitor.

Expected LCD Output

System Initialization

Smart Building

System Ready

RFID Authentication

Scan RFID Card

RFID Verified

RFID Verified

Place Finger

Fingerprint Verified

Finger Verified

Door Opening

Password Authentication

Enter Password

Door Opened

Welcome User

Door Open

Door Closed

Door Closed

System Locked

Motion Detected

Motion Detected

Checking...

Security Alert

Access Denied

Alarm Active

Cloud Upload

Uploading...

Success

Expected UART Output

Initialization

LCD Initialized

Display Ready

Display Updated

LCD Message Updated

LCD Error

LCD Communication Failed

Check LCD Connection

LED Indication

LCD Working Normally

 Green LED : ON

 Red LED : OFF

LCD Updated

 Green LED : Blinking

 Red LED : OFF

LCD Error

 Green LED : OFF

 Red LED : ON

Relay Module

Explanation

The **Relay Module** is used to control the **Electromagnetic Door Lock** in the Smart Building Security, Access Control and IoT Monitoring System. It is connected to the **LPC2129 ARM7**

Microcontroller through a GPIO pin. After successful **RFID authentication, fingerprint verification, and password validation (if enabled)**, the LPC2129 energizes the relay to unlock the door. If authentication fails, the relay remains OFF, keeping the door locked for security.

Expected UART Output

Initialization

Relay Module Initialized

Relay Ready

Relay ON (Door Unlock)

Authentication Successful

Relay ON

Door Unlock Command Executed

Relay OFF (Door Lock)

Door Closed

Relay OFF

Door Locked

Authentication Failed

Authentication Failed

Relay OFF

Access Denied

Expected LCD Output

Initialization

Relay Ready

Relay ON

Door Unlock

Relay ON

Relay OFF

Door Locked

Relay OFF

Authentication Failed

Access Denied

Door Locked

LED Indication

Relay OFF (Door Locked)

 Green LED : OFF

 Red LED : ON

Relay ON (Door Unlocked)

 Green LED : ON

 Red LED : OFF

Authentication Failed

 Green LED : OFF

 Red LED : ON

 Buzzer : ON

Electromagnetic Door Lock

Explanation

The **Electromagnetic Door Lock** is used to **secure the building entrance** in the Smart Building Security, Access Control and IoT Monitoring System. It is controlled by the **LPC2129 ARM7 Microcontroller** through the **Relay Module**. After successful **RFID authentication, fingerprint verification, and password validation (if enabled)**, the LPC2129 energizes the relay to unlock the electromagnetic door lock, allowing the user to enter. If authentication fails, the door remains locked, preventing unauthorized access.

Expected UART Output

Initialization

Electromagnetic Door Lock Ready

Door Locked

Door Unlock

Authentication Successful

Door Unlocking...

Access Granted

Door Lock

Door Closed

Door Locked

System Secured

Unauthorized Access

Authentication Failed

Door Remains Locked

Access Denied

Expected LCD Output

Initialization

Door Locked

System Ready

Door Unlock

Welcome User

Door Open

Door Lock

Door Closed

Locked

Unauthorized Access

Access Denied

Door Locked

LED Indication

Door Locked

 Green LED : OFF

 Red LED : ON

Door Unlocked

 Green LED : ON

 Red LED : OFF

Unauthorized Access

 Green LED : OFF

 Red LED : ON

 **Buzzer : ON**

Servo Motor

Explanation

The **Servo Motor** is used to **automatically open and close the door** in the Smart Building Security, Access Control and IoT Monitoring System. It is controlled by the **LPC2129 ARM7 Microcontroller** using a PWM signal. After successful **RFID authentication, fingerprint verification, and password validation (if enabled)**, the LPC2129 rotates the servo motor to open the door. After a few seconds, the servo returns to its original position to close the door automatically, ensuring secure access.

Expected UART Output

Initialization

Servo Motor Initialized

Door Control Ready

Door Opening

Authentication Successful

Servo Rotating to 90°

Door Opened

Door Closing

Closing Door...

Servo Rotating to 0°

Door Closed

Authentication Failed

Authentication Failed

Servo Not Activated

Door Remains Closed

Expected LCD Output

Initialization

Servo Ready

Door Opening

Door Opening

Please Enter

Door Closing

Door Closing

Thank You

Authentication Failed

Access Denied

Door Closed

LED Indication

Servo Idle (Door Closed)

 Green LED : OFF

 Red LED : ON

Servo Operating (Door Opening)

 Green LED : Blinking

 Red LED : OFF

Door Fully Open

 Green LED : ON

 Red LED : OFF

Authentication Failed

 Green LED : OFF

 Red LED : ON

 **Buzzer : ON**

LED Indicators and Buzzer

Explanation

The **LED Indicators** and **Buzzer** are used to provide **visual and audio indications** about the system status. They are connected to the **LPC2129 ARM7 Microcontroller** through GPIO pins. The **Green LED** indicates successful system operation or user authentication, while the **Red LED** and **Buzzer** indicate authentication failure, unauthorized access, door breach, or security alerts. These indicators help users and security personnel quickly understand the current status of the system.

Expected UART Output

Initialization

LED Indicators Initialized

Buzzer Initialized

System Status Indicators Ready

Authentication Successful

Authentication Successful

Green LED ON

Buzzer OFF

Authentication Failed

Authentication Failed

Red LED ON

Buzzer ON

Security Alert

Unauthorized Access Detected

Red LED ON

Buzzer Activated

Security Alert Generated

Expected LCD Output

Initialization

System Ready

Authentication Successful

Access Granted

Welcome User

Authentication Failed

Access Denied

Try Again

Security Alert

Security Alert

Check Door

LED & Buzzer Indication

System Ready

 **Green LED : ON**

 **Red LED : OFF**

 **Buzzer : OFF**

Authentication Successful

 **Green LED : ON**

 **Red LED : OFF**

 **Buzzer : OFF**

Authentication Failed

 **Green LED : OFF**

 **Red LED : ON**

 **Buzzer : ON (2 Seconds)**

Unauthorized Access / Door Breach

 **Green LED** : OFF

 **Red LED** : Blinking

 **Buzzer** : Continuous ON Until Reset

Motion Detected (Monitoring Mode)

 **Green LED** : Blinking

 **Red LED** : OFF

 **Buzzer** : OFF

Communication Interfaces

UART is used for communication with the ESP-01 Wi-Fi module and Arduino UNO. SPI is used for the RC522 RFID Reader and SD Card Module. I²C is used for the DS1307 RTC and AT24C256 EEPROM. CAN communication is implemented through the MCP2551 transceiver. GPIO pins interface with the keypad, PIR sensor, magnetic door sensor, user switches, relay, servo motor, LEDs, buzzer and LCD.

Applications

- Corporate Offices
- Industrial Plants
- Hospitals
- Educational Institutions
- Apartments and Residential Buildings
- Banks

- Government Offices
- Laboratories
- Data Centers
- Research Organizations

14. Advantages

- Provides multi-level authentication.
- Improves building security.
- Automatic door control.
- Real-time event logging.
- Cloud-based monitoring.
- Supports CAN networking.
- Scalable and modular architecture.
- Reliable and easy to maintain.

15. Future Scope

Future enhancements may include face recognition, mobile application control, AI-based visitor management, voice authentication, biometric attendance integration, cloud analytics dashboards, emergency evacuation support, and complete smart building automation.

16. Conclusion

The Smart Building Security, Access Control and IoT Monitoring System demonstrates the integration of embedded systems, IoT, industrial communication and multiple authentication techniques into a single intelligent security solution. By combining RFID, fingerprint verification, password authentication, cloud connectivity, data logging and CAN

communication, the project provides secure, reliable and scalable access control for modern buildings.

17. Technical Interview Questions

1. Why is LPC2129 used in this project?
2. Why is Arduino required for the fingerprint sensor?
3. Explain the authentication sequence.
4. What is the purpose of the PIR sensor?
5. What is the role of the magnetic door sensor?
6. Why is RTC used?
7. What is the purpose of EEPROM and SD Card?
8. Explain the role of the ESP-01 module.
9. Why is MCP2551 used?
10. How does the system respond to unauthorized access?

18. Final Expected UART Output

Case-1: Authorized User (Access Granted)

=====

Smart Building Security System

=====

System Initializing...

RC522 Ready

Fingerprint Sensor Ready

RTC Ready

EEPROM Ready

SD Card Ready

ESP-01 Connected

CAN Network Ready

Waiting for RFID Card...

RFID Card Detected

RFID Authentication Successful

Enter Password...

Password Verified

Place Finger...

Fingerprint Verified

Access Granted

Relay ON

Door Unlocking...

Servo Rotating to 90°

Door Opened

Door Status : OPEN

Security Log Saved

Cloud Upload Successful

CAN Message Sent

Welcome User

Case-2: Unauthorized User (Access Denied)

=====

Smart Building Security System

=====

System Initializing...

RC522 Ready

Fingerprint Sensor Ready

RTC Ready

EEPROM Ready

SD Card Ready

ESP-01 Connected

CAN Network Ready

Waiting for RFID Card...

RFID Card Detected

RFID Authentication Failed

OR

Password Incorrect

OR

Fingerprint Not Matched

Access Denied

Relay OFF

Door Locked

Red LED ON

Buzzer Activated

Security Log Saved

Cloud Alert Sent

CAN Alert Message Sent

Unauthorized Access Detected

All the best...

